Essays

|!|BCEEC0250000001|!|

Describe and explain how the different components of blood are involved in body defence against pathogens. (12 marks)

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Blood clotting: (Max. 3)

- Blood platelets and fibrinogen in blood plasma are involved in blood clotting. (1)
- When a blood vessel is damaged, blood platelets are attracted to the wound by chemicals released from the vessel wall. (1)
- The blood platelets and the damaged cells at the wound release substances that start a series of reactions. These reactions eventually cause fibrinogen to convert into fibrin. (1)
- The fibrin threads form a network which traps red blood cells, blood platelets and white blood cells, forming a blood clot. (1)

Phagocytosis: (Max. 2)

- Phagocytosis involves white blood cells called phagocytes. (1)
- When a phagocyte encounter a pathogen, it encloses the pathogen in a vacuole. (1)
- Lysosome in the phagocyte fuses with the vacuole containing the pathogen and lysozyme is released to digest the pathogen. (1)

Immune responses: (Max. 4)

- Specific immune responses involve lymphocytes: T cells and B cells. (1)
- When an antigen of a pathogen binds to a B cell with the matching receptor, the B cell becomes activated and divides and differentiate into plasma cells and memory B cells. (1)
- Plasma cells secrete antibodies into blood, tissue fluid and lymph to act against the invading pathogen. (1)
- Memory B cells ‘memorize’ the antigen. They are important for the secondary response. (1)
- When a helper T cell detects an antigen, it becomes activated and releases chemicals called lymphokines. (1)
- Lymphokines activate other T cells to divide and differentiate into killer T cells and memory T cells. (1)
- Killer T cells bind to the cells infected by pathogens or cancer cells and destroy them directly. (1)
- Memory T cells ‘memorize’ the antigen. They are important for the secondary response. (1)

Communication (Max. 3)

(12 marks)

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Skills assessed	Marks
Understanding basic concepts	9
Communication	3

|!|BCEEC0250000002|!|

Describe and give examples about how humans acquire immunity actively and passively. Note that details of the actions of B cells and T cells are not required. (12 marks)

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Active immunity: (Max. 2)

- Introduction of antigens into the human body leads to immune responses (1)
- Antibody production by plasma cells is stimulated by the antigens (1)
- Formation of memory cells (1)

Natural active immunity: (1)

- Infection by pathogens, e.g. bacteria, fungi, virus, etc.

Artificial active immunity: (1)

- Vaccination; injection of weakened or killed pathogen, or its antigen

Passive immunity: (Max. 3)

- The body receives ready-made antibodies (1)
- Providing immediate protection (1)
- Immunity is short-term as the antibodies are used up (1)
- No memory cells are formed (1)

Natural passive immunity: (1)

- Antibodies transfer from a mother to a foetus across the placenta (1)
- Antibodies transfer from a mother to an infant through breast feeding (1)

Artificial passive immunity: (1)

- Injection of ready-made antibodies, e.g. serum containing high concentration of antibodies (1)

Communication (Max. 3)

(12 marks)

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Skills assessed	Marks
Understanding basic concepts	9
Communication	3

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